

# Phylogenetics

The Big Recap

RL-V3 MPP

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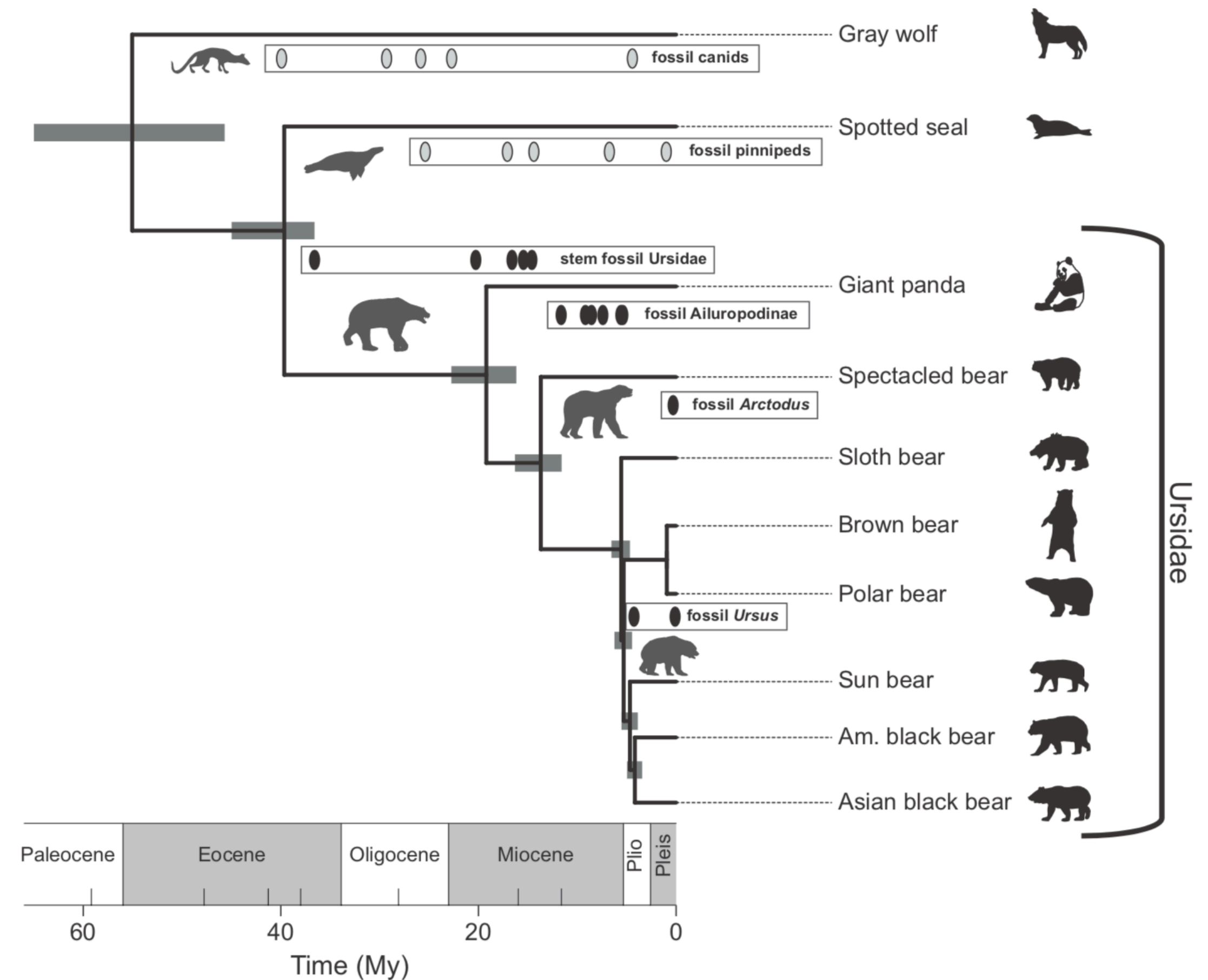
14.04.2026



# Today

- **Recap**

- Bayesian phylogenetic dating
- The fossilised birth-death process
- Total-evidence dating



# Recap: Bayesian phylogenetic dating

# We use a Bayesian framework

$$P(\text{model} \mid \text{data}) = \frac{P(\text{data} \mid \text{model}) P(\text{model})}{P(\text{data})}$$

Diagram illustrating the Bayesian framework components:

- likelihood**: Points to  $P(\text{data} \mid \text{model})$
- priors**: Points to  $P(\text{model})$
- posterior**: Points to  $P(\text{model} \mid \text{data})$
- marginal probability of the data**: Points to  $P(\text{data})$

# Bayesian divergence time estimation

## The data

and / or

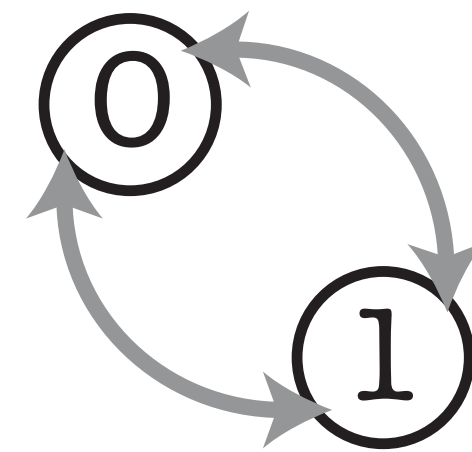
0101... ATTG...  
1101... TTGC...  
0100... ATTC...



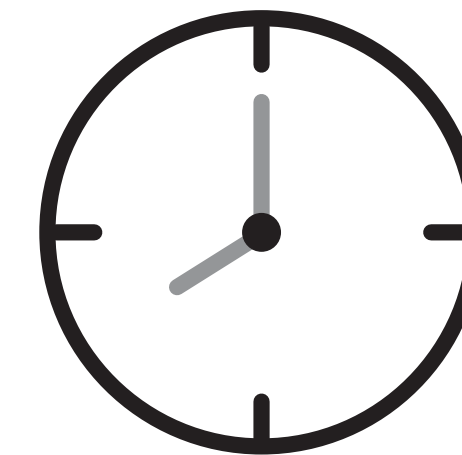
phylogenetics  
characters

sample  
ages

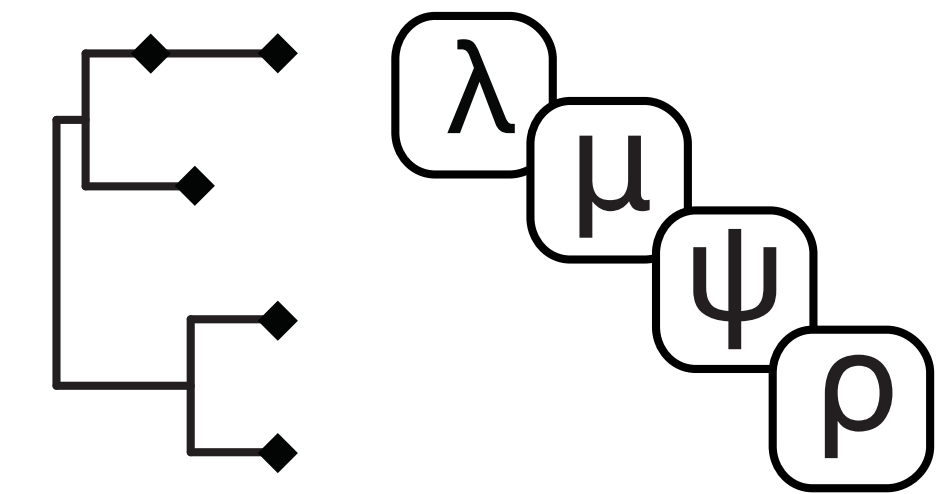
## 3 model components



substitution  
model

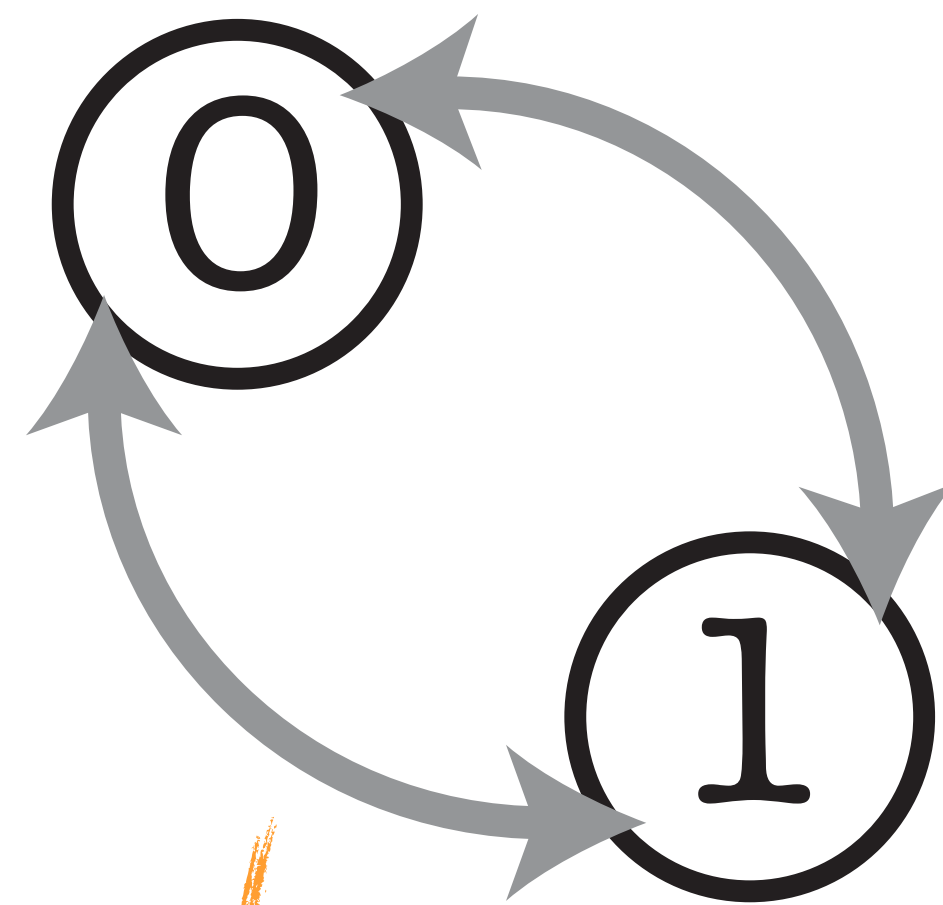


clock  
model

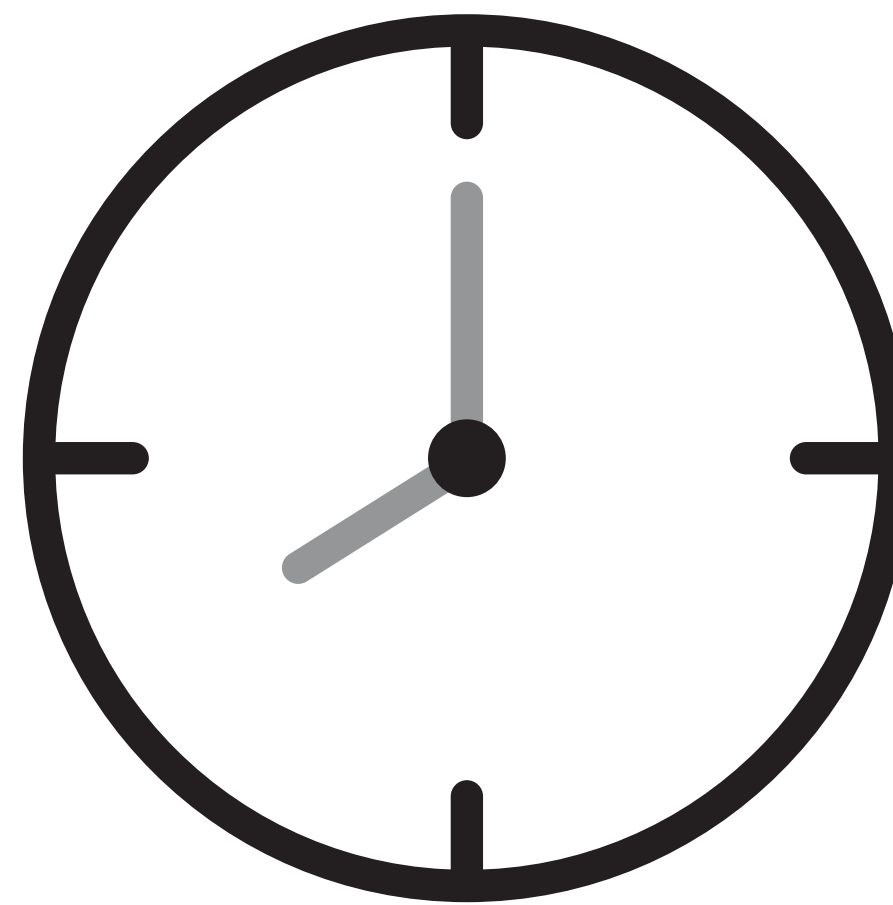


tree and tree  
model

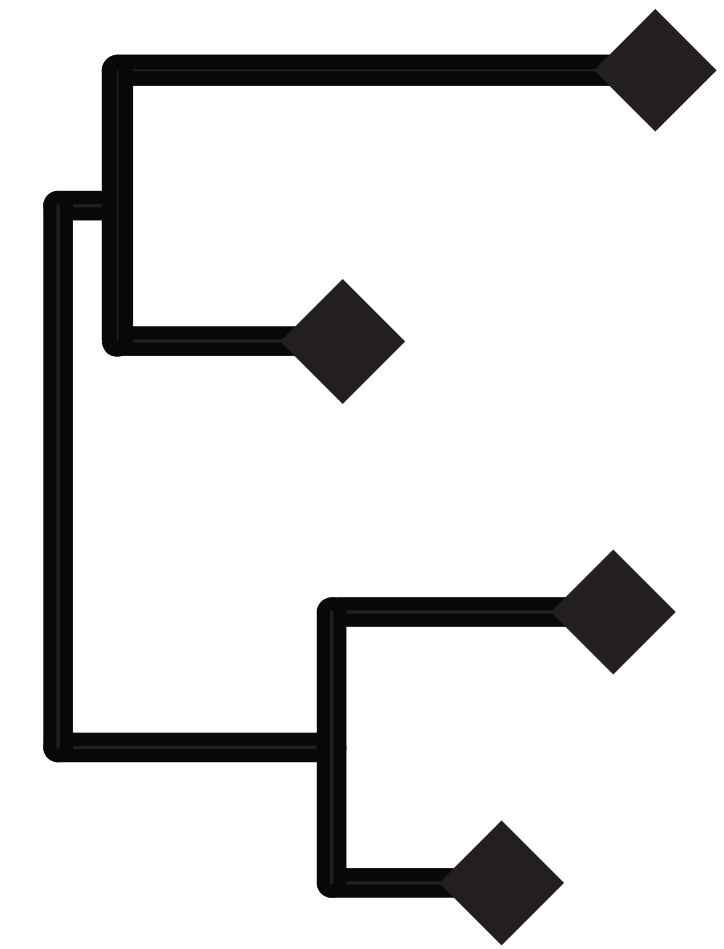
substitution model



clock model

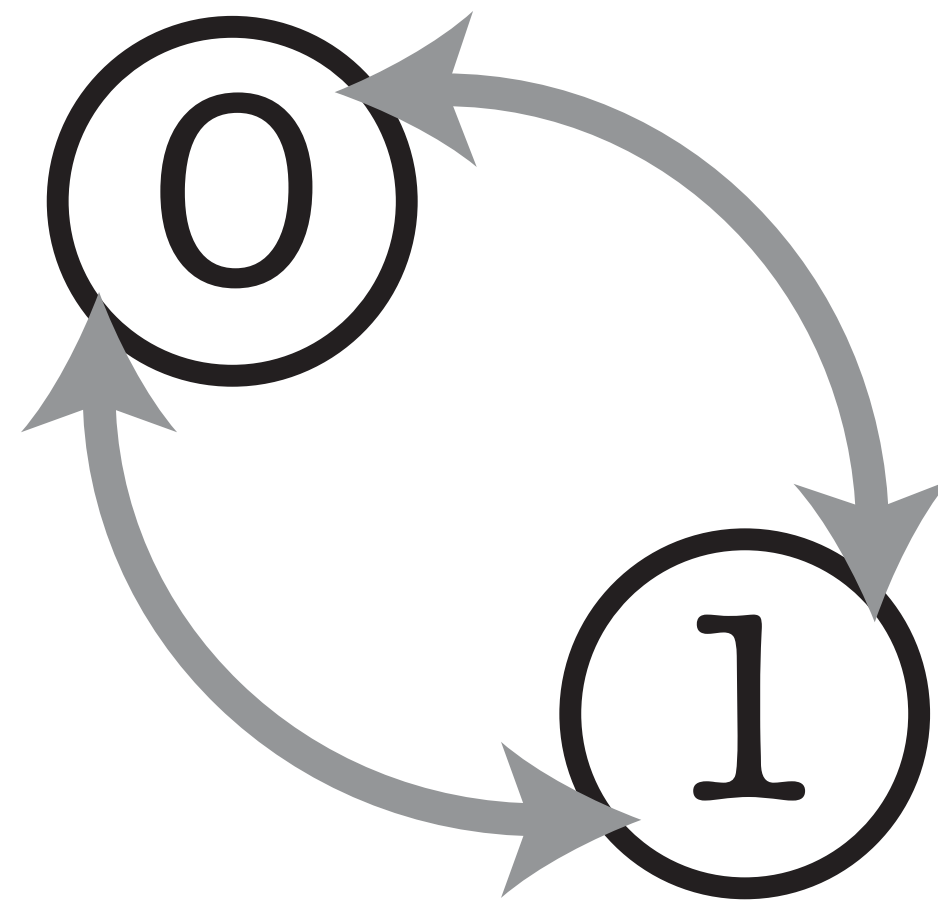


tree model

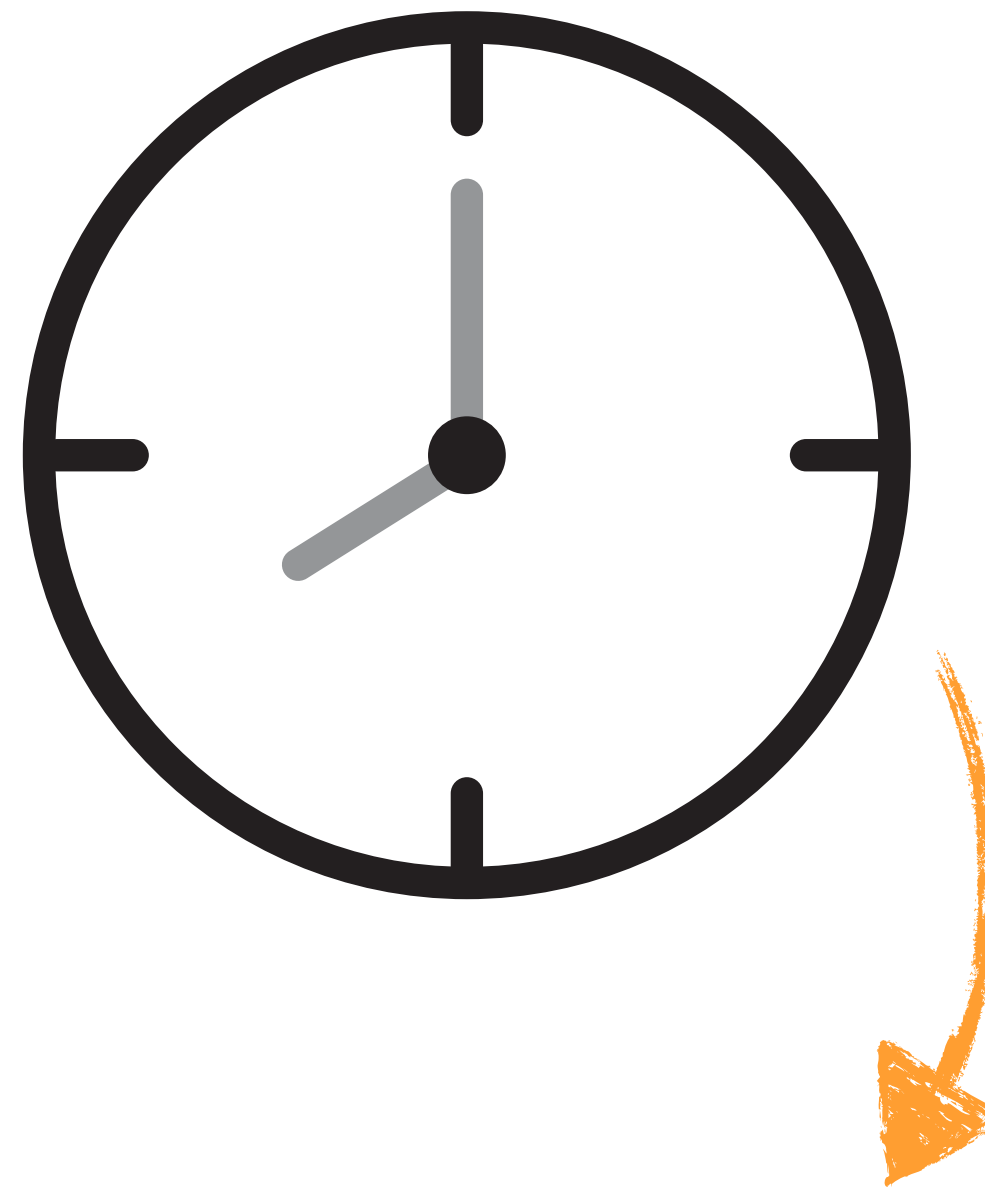


How likely are we to observe a change between character states? e.g.,  $A \rightarrow T$

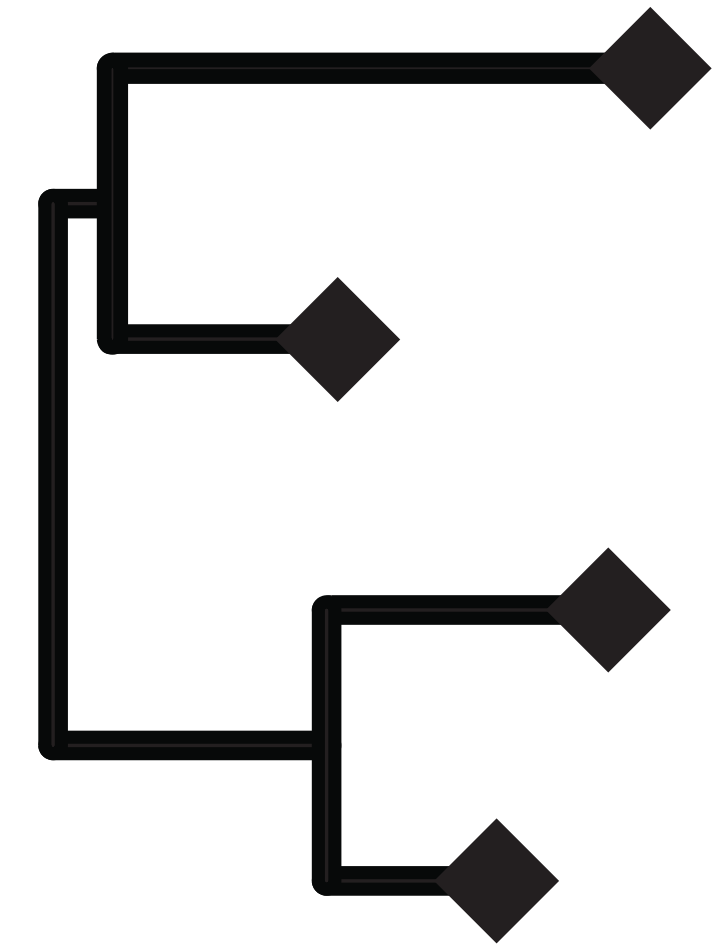
substitution model



clock model

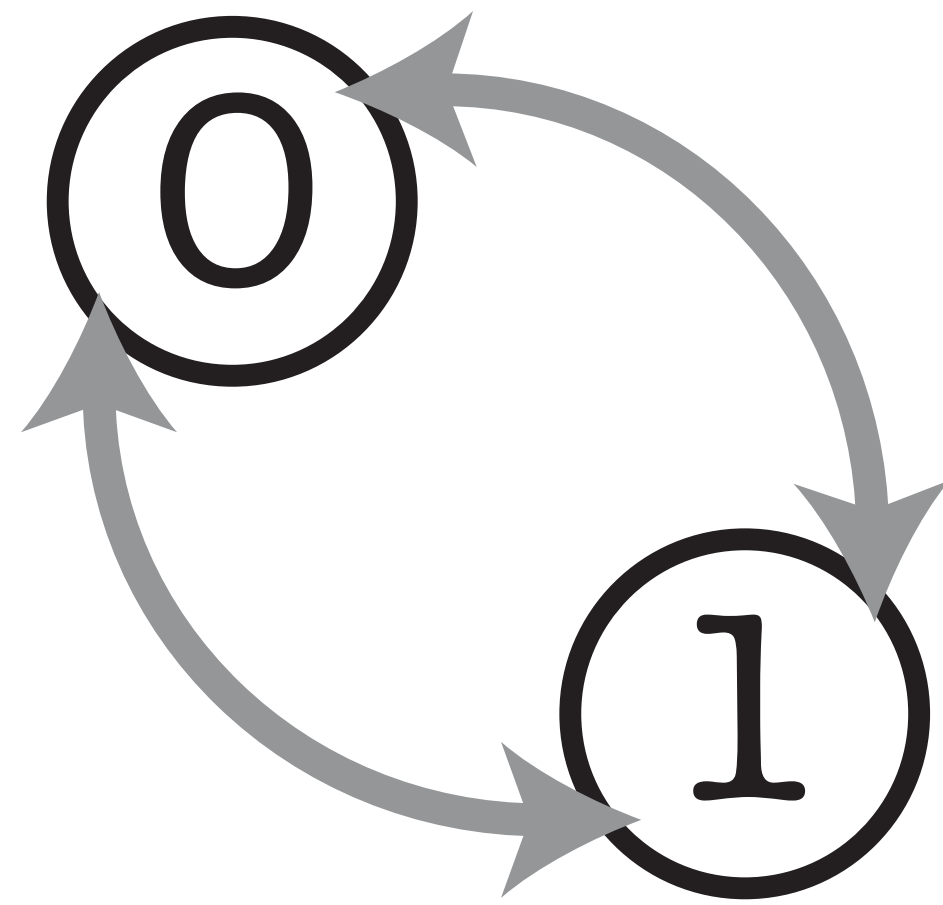


tree model

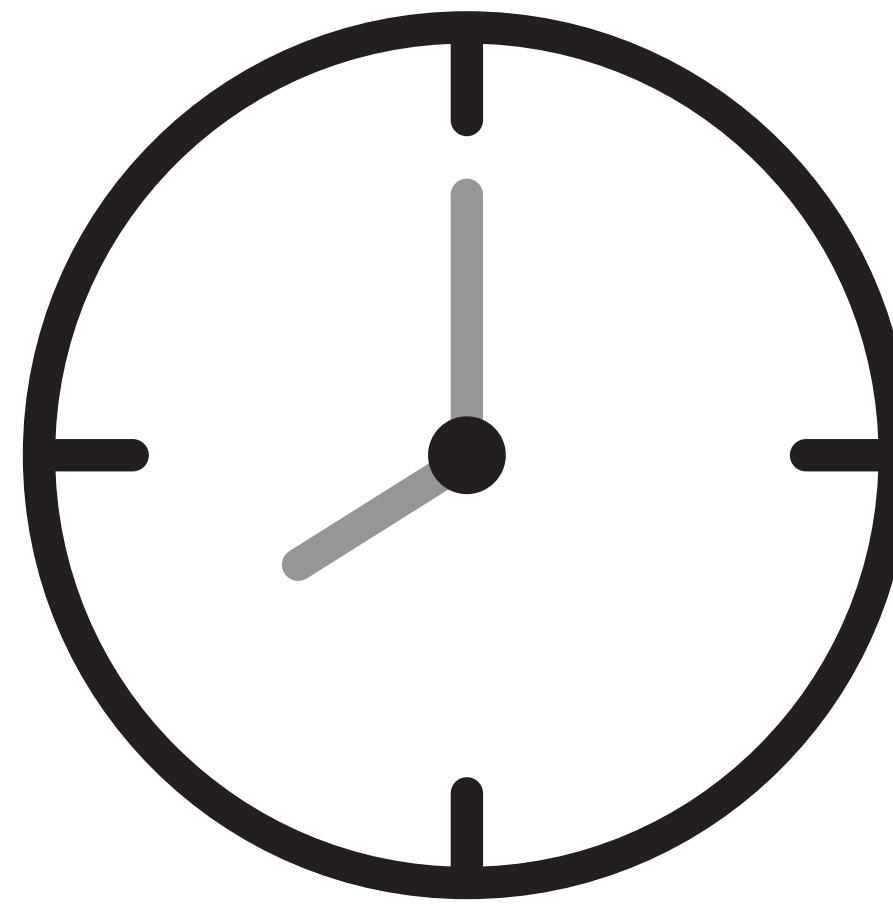


How have rates of evolution varied  
(or not) across the tree?

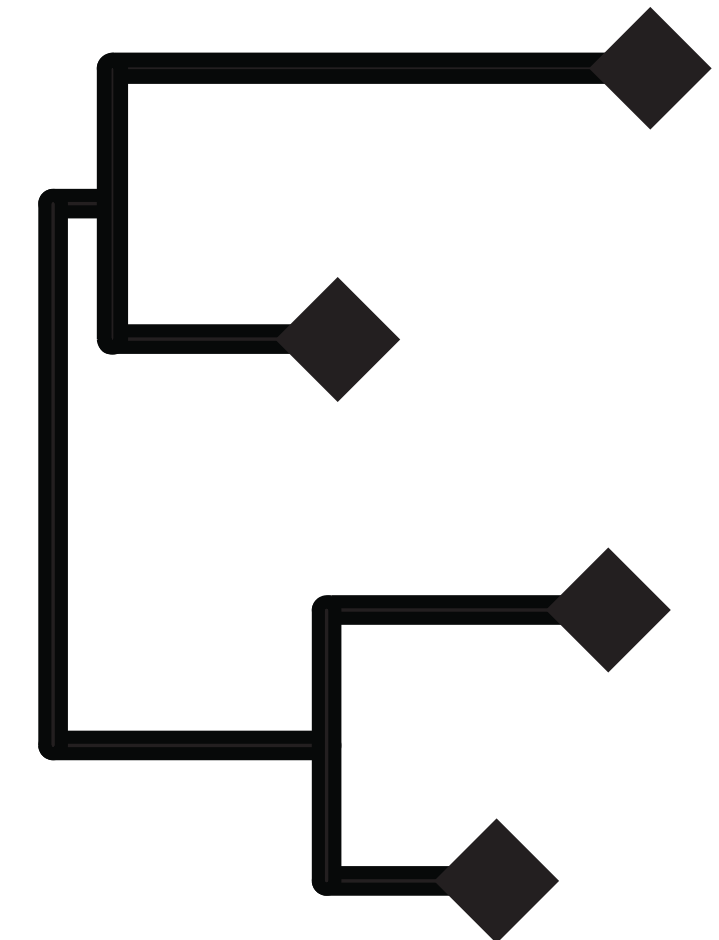
substitution model



clock model



tree model



How have species originated, gone extinct and been sampled through time?

# Bayesian divergence time estimation

posterior

$$P\left(\begin{array}{c} \text{tree} \\ \lambda, \mu, \psi, \rho \\ \text{state} \\ \text{clock} \end{array} \mid \begin{array}{c} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \text{data}\right) =$$

likelihood

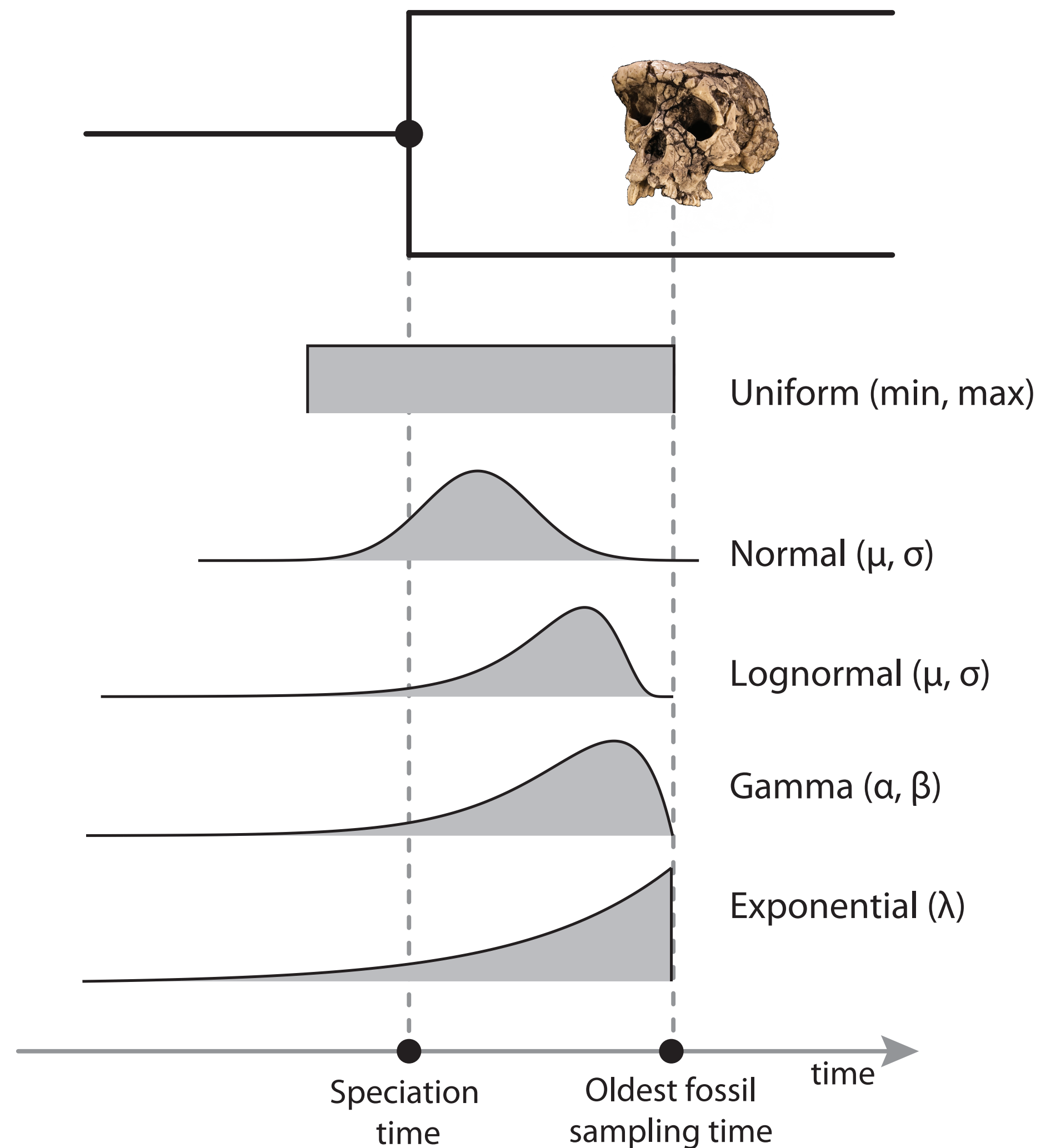
probability of the  
time tree

priors

$$\frac{P\left(\begin{array}{c} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \mid \begin{array}{c} \text{tree} \\ \lambda, \mu, \psi, \rho \\ \text{state} \\ \text{clock} \end{array}\right) P\left(\begin{array}{c} \text{tree} \\ \lambda, \mu, \psi, \rho \\ \text{state} \end{array} \mid \text{data}\right) P\left(\begin{array}{c} \lambda, \mu, \psi, \rho \end{array}\right) P\left(\begin{array}{c} \text{state} \end{array}\right) P\left(\begin{array}{c} \text{clock} \end{array}\right)}{P\left(\begin{array}{c} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \mid \text{data}\right)}$$

marginal pr of the data

# Recap: Node dating

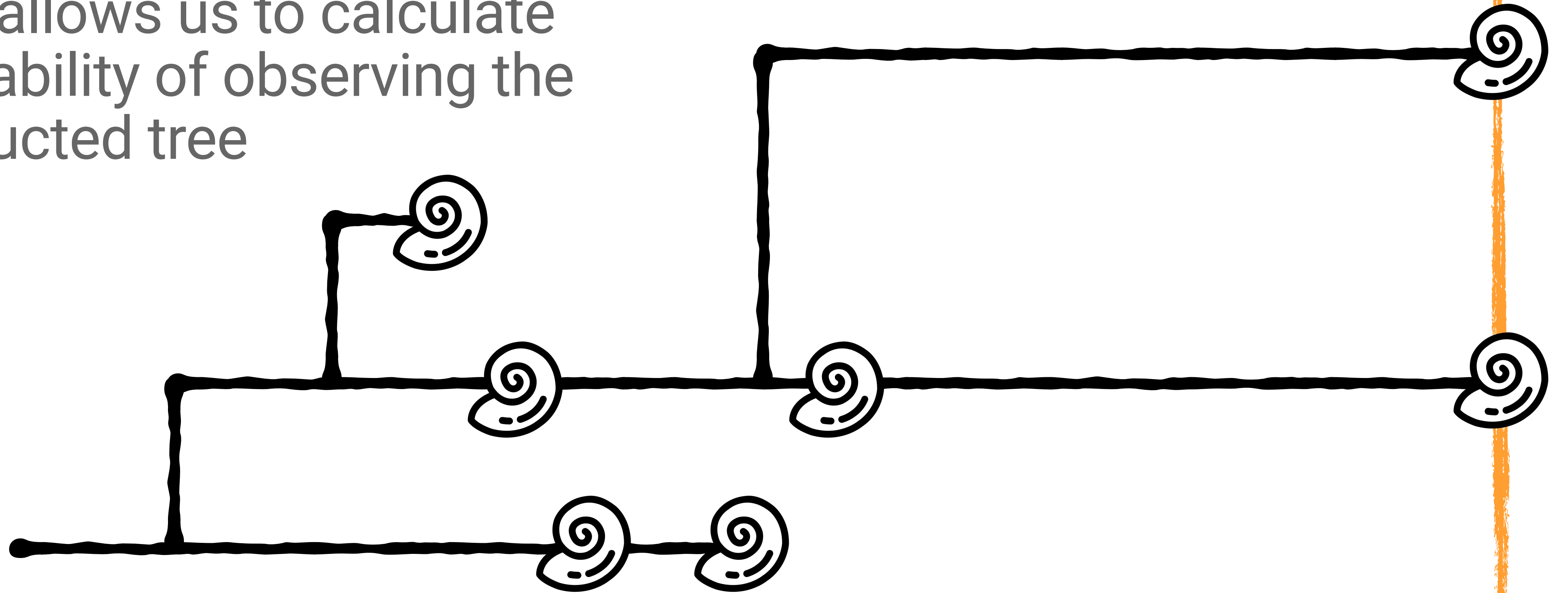


The tree includes extant (living) species only

We can use a **calibration density** to constrain internal node ages

We typically use a **birth-death process** model to describe the tree generating process

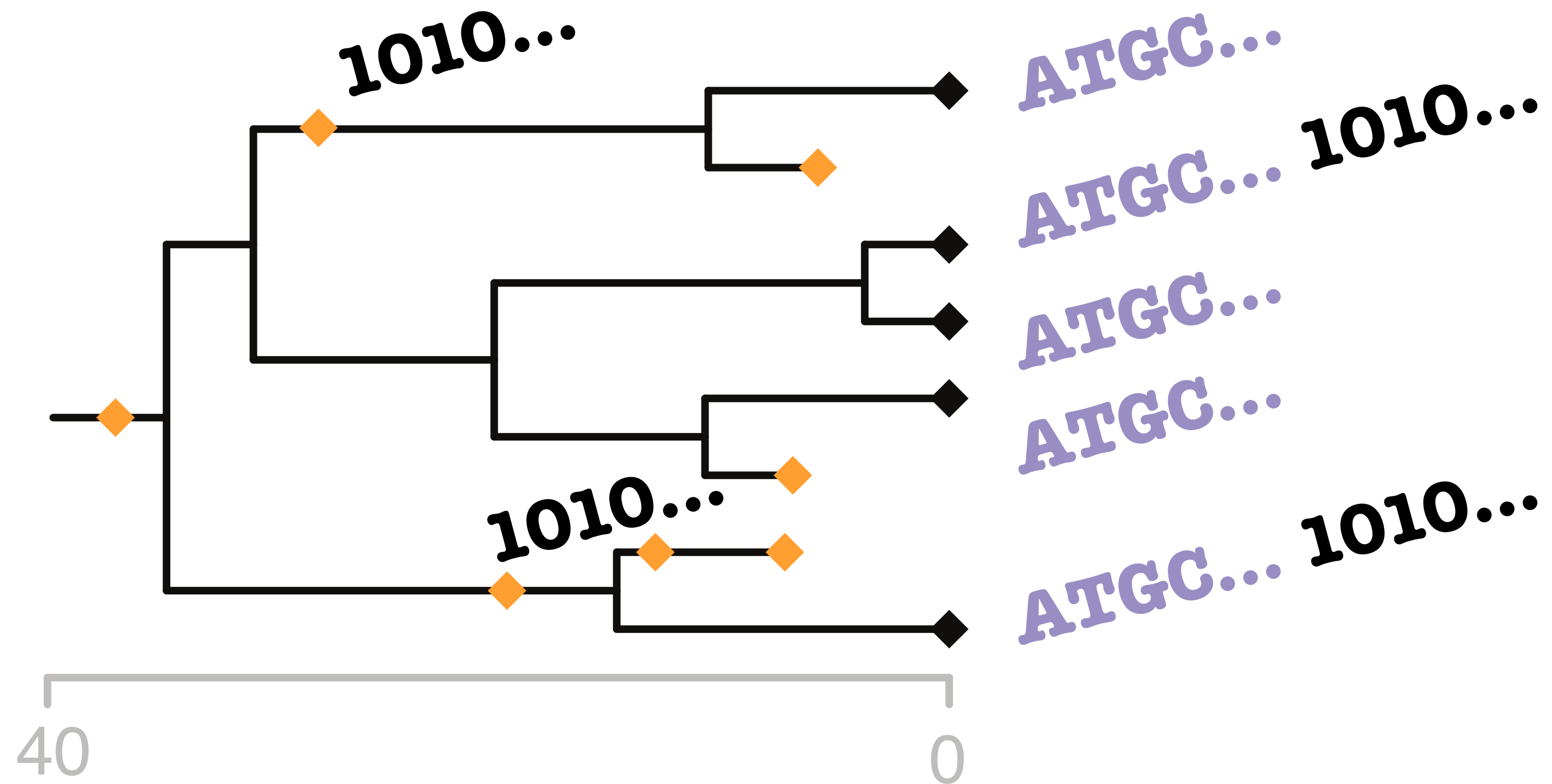
The **fossilised birth-death (FBD) process** allows us to calculate the probability of observing the reconstructed tree



$$P(\mathcal{E} | \text{FBD}(\lambda, \mu, \psi, \rho))$$

*Sampling-through-time in birth-death trees. Stadler. (2010)  
First implemented: Heath et al. (2014) and Gavryushkina et al. (2014)*

# Recap: “total-evidence” dating



We have DNA for living species. We have morphology for living *and* fossil species

Fossils can be positioned on the basis of morphology  
→ accounts for uncertainty in fossil placement

Fossils can be incorporated via **taxonomy** or **character data** (total-evidence)

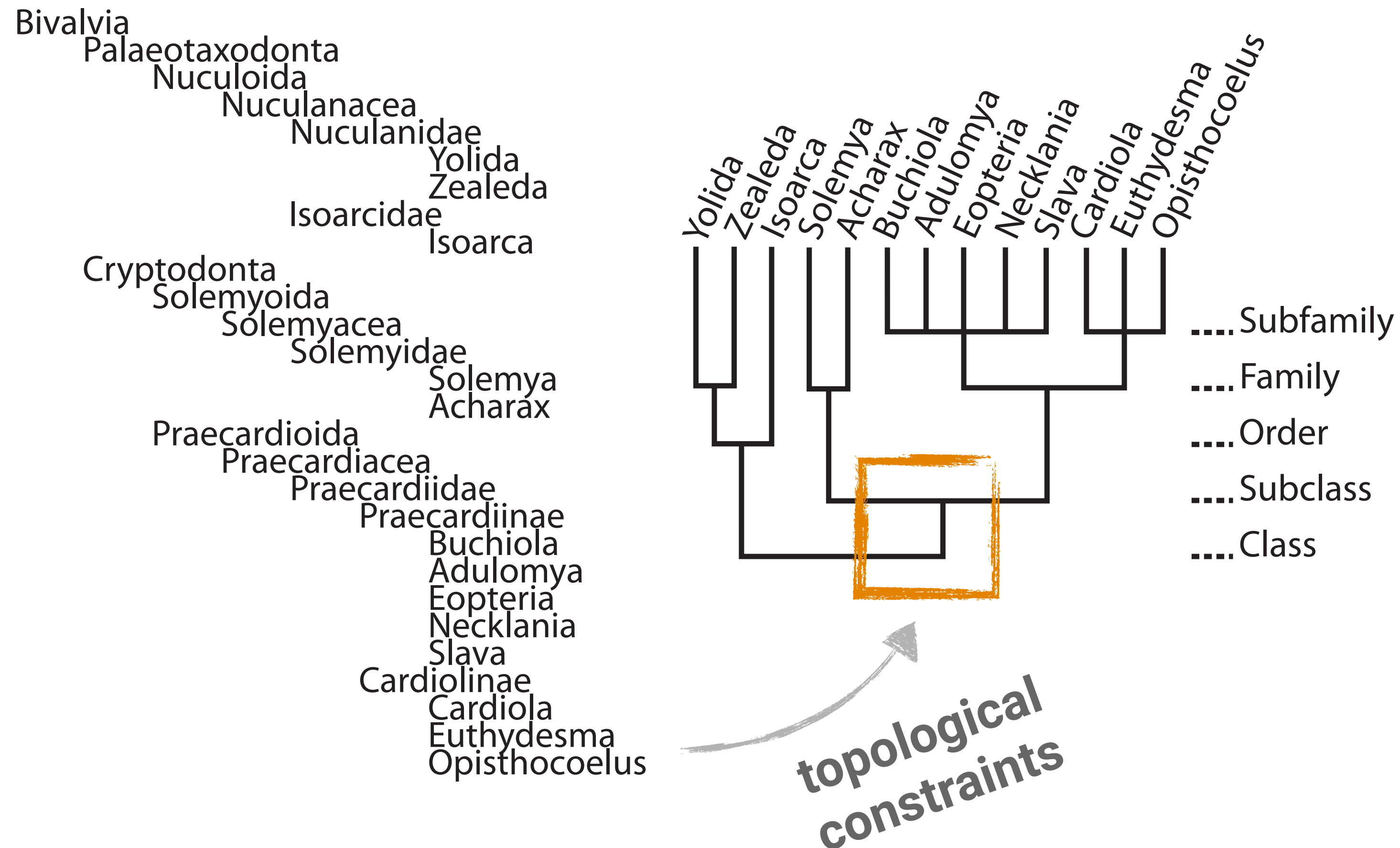
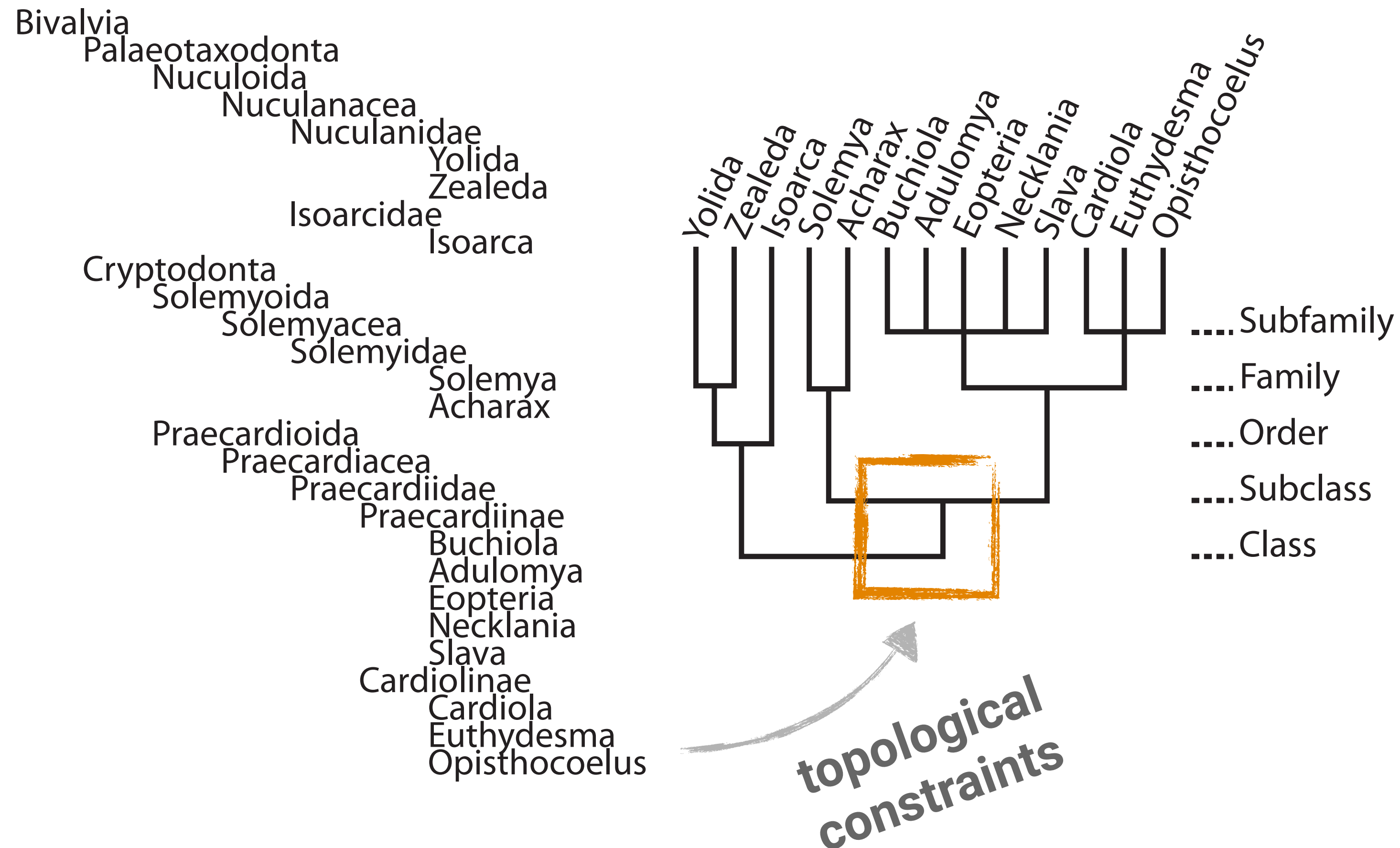


Image source Soul & Friedman (2015)

Fossils can be incorporated via **taxonomy** or **character data** (total-evidence)



**A**T**A**T...

# TCAC...

????...

OR

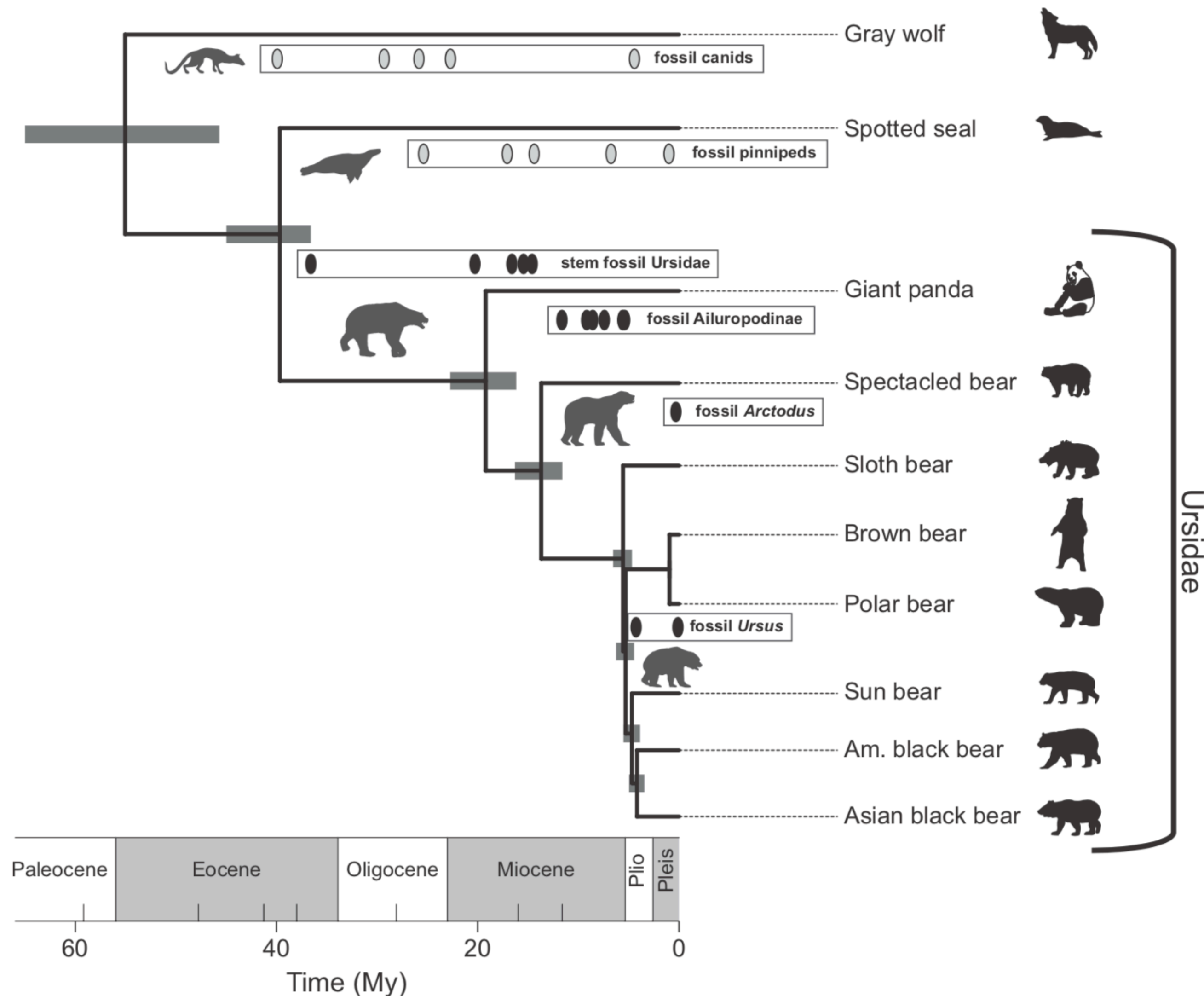
**1001...**

1101...

0100...

Image source Soul & Friedman (2015)

# Time calibrated tree of living and fossil bears



First application of the FBD model

Fossils are incorporated via constraints, not character data. Their precise placement can not be inferred, but this uncertainty will be reflected in the posterior

# What does it mean to run the model “under the prior”?

posterior

$$P\left(\begin{array}{c} \text{tree} \\ \lambda, \mu, \psi, \rho \end{array} \begin{array}{c} \text{cyc} \\ 0, 1 \end{array} \text{clock} \mid \begin{array}{c} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \text{spiral} \right) =$$

likelihood

probability of the  
time tree

priors

~~$$P\left(\begin{array}{c} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \mid \begin{array}{c} \text{tree} \\ \lambda, \mu, \psi, \rho \end{array} \begin{array}{c} \text{cyc} \\ 0, 1 \end{array} \text{clock}\right) P\left(\begin{array}{c} \text{tree} \\ \lambda, \mu, \psi, \rho \end{array} \mid \text{spiral}\right) P\left(\begin{array}{c} \lambda, \mu, \psi, \rho \end{array}\right) P\left(\begin{array}{c} \text{cyc} \\ 0, 1 \end{array}\right) P(\text{clock})$$~~

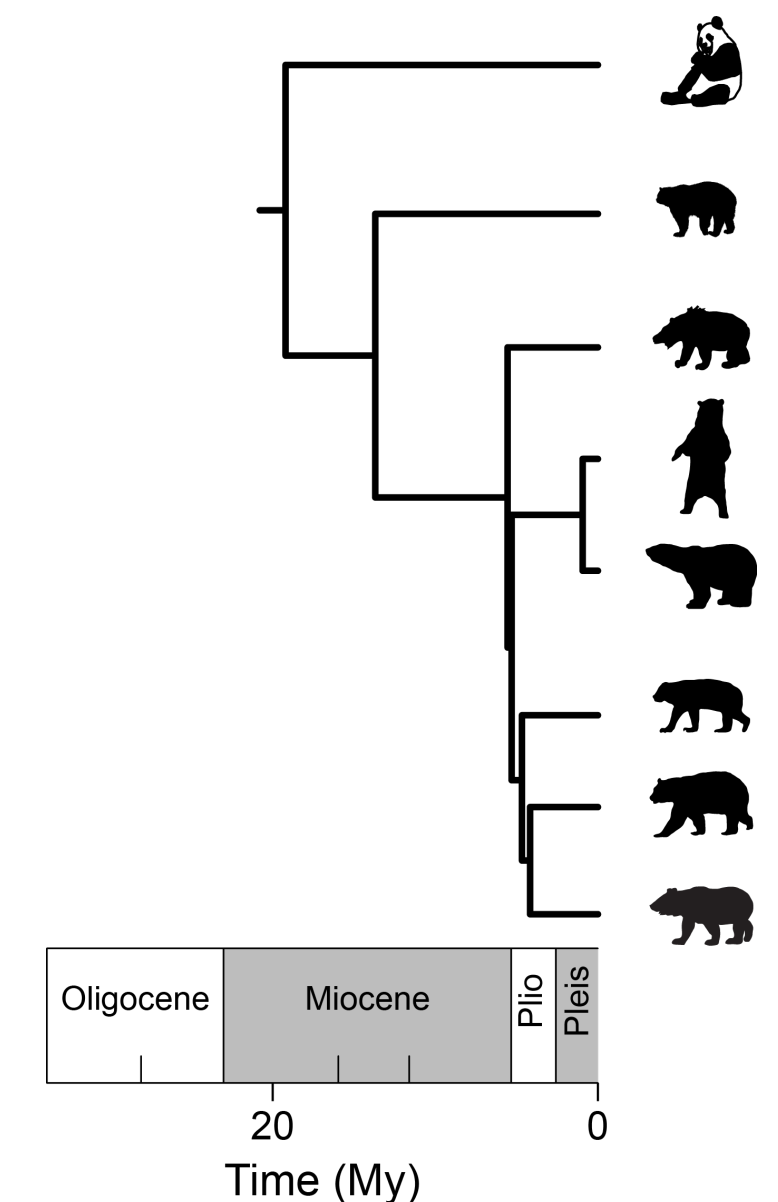
$$P\left(\begin{array}{c} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \mid \text{spiral}\right)$$

Ignore this part! i.e.,  
the sequence data

marginal pr of the data

# Exercise (review set up and output)

- 2 gene sequences (cytb, irbp) for **extant bears only**
- fossil ages
- no character data for the fossils



# Exercise

- molecular sequences for living ferns
- fossil ages
- morphology for living and fossil species

